Week 12 Exercises

Research Paper. Continue to make progress on your research paper. The next draft is due Tuesday of Thanksgiving week. This draft should be close to finalized, so that I can make suggestions to help polish the paper.

Workshop. Finalize your materials.

Rehearsals:

- ~~*Cox.* Monday, November 10 at 9am (to about 10am).~~
- *Unordered* and *ordered.* Friday, November 14 at 12pm (to about 2pm).

Workshops: I've solicited RIBC students and sent around the (still active) registration sheet.

- *Unordered.* Tuesday, November 18 from 1pm to 2pm.
- *Ordered.* Wednesday, November 19 from 1pm to 2pm.
- *Cox.* Friday, November 21 from 11am to 12noon.

Exercise 1 Simulate and recover; log-normal with censoring

Do the simulate and recover exercise for a log-normal duration model *with censoring*. You may use any `comparison()` you like.

> ⚠ Warning
>
> For the log-normal, we have $\text{median}(y \mid X) = \exp(X\beta)$ and $E(y \mid X) = \exp\left(X\beta + \frac{\sigma^2}{2}\right)$. You can find these identities listed in the back of most probability textbooks.
> When writing the solution, I realized `predict()` returns the *median* of Y for `survreg(..., dist = "lognormal")`. This means that `predictions()` returns medians and `comparisons()` returns differences in medians.
> But there's also the {flexsurvreg} package. I tested it further and realized `predict()` returns the *mean* of Y for `flexsurvreg(..., dist = "lognormal")`. This means that `predictions()` returns means and `comparisons()` returns differences in means for

`flexsurvreg()` models.
This is an example of software doing things you don't realize or expect, but you can catch with a simulate and recover experiment.

```
## --- simulate a fake data set: log-normal duration model with censoring ---

# set sample size
n <- 1000  # large so that CI is not super wide

# create explanatory variables
## numeric variable
set.seed(123)
x1 <- rnorm(n, 0, 0.5)  # sd = 0.5 is my preferred scale for interpretation
## qualitative variable
x2 <- sample(c("Label 1", "Label 2"),
             size = n,
             replace = TRUE,
             prob = c(0.75, 0.25))  # about 75% will be "Label 1"

# create coefficients
b0 <- 0
b1 <- 0.6
b2 <- -0.4
sigma <- 0.4

# simulate event times from LogNormal with meanlog = eta and sdlog = sigma
eta <- b0 + b1 * x1 + b2 * (x2 == "Label 2")
t_unobs <- rlnorm(n, meanlog = eta, sdlog = sigma)


c <- 1 # censored here; about 25% will be censored

# observed time and censoring indicator
time <- ifelse(t_unobs > c, c, t_unobs)
status <- as.integer(t_unobs < c)  # 1 = event observed, 0 = censored

# combine into data frame
data <- data.frame(time, status, x1, x2) |>
  glimpse()
```

```
Rows: 1,000
Columns: 4
$ time   <dbl> 0.6086345, 0.8253431, 1.0000000, 1.0000000, 1.0000000, 1.000000~
$ status <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, ~
$ x1     <dbl> -0.28023782, -0.11508874, 0.77935416, 0.03525420, 0.06464387, 0~
$ x2     <chr> "Label 1", "Label 1", "Label 1", "Label 1", "Label 1", "Label 1~
```

```
# for log-normal
# - E(T | x) = mean(T | x) = exp(eta + sigma^2 / 2)
# - median(T | x) = exp(eta)

eta_hi <- b0 + b1 * max(x1) + b2 * 1  # x2 == "Label 2"
eta_lo <- b0 + b1 * min(x1) + b2 * 1

# medians
median_hi <- exp(eta_hi)
median_lo <- exp(eta_lo)

# means
mean_hi <- exp(eta_hi + (sigma^2) / 2)
mean_lo <- exp(eta_lo + (sigma^2) / 2)

# predict() returns medians (I learned!), so predictions() and comparisons() return medians
truth_fd <- median_hi - median_lo
truth_fd
```

```
[1] 1.483823
```

```
# --- recover the quantity of interest ---

# load packages
library(survival)
```

```
Attaching package: 'survival'

The following object is masked from 'package:brms':

    kidney
```

```r
library(flexsurv)
library(marginaleffects)

# fit log-normal AFT with censoring
fit <- survreg(Surv(time, status) ~ x1 + x2, data = data, dist = "lognormal")

summary(fit) # note that log(.4) = -0.9162907
```

```
Call:
survreg(formula = Surv(time, status) ~ x1 + x2, data = data,
    dist = "lognormal")
               Value Std. Error      z      p
(Intercept) -0.00304    0.01778  -0.17   0.86
x1           0.55779    0.03058  18.24 <2e-16
x2Label 2   -0.39787    0.03103 -12.82 <2e-16
Log(scale)  -0.93861    0.03168 -29.63 <2e-16

Scale= 0.391

Log Normal distribution
Loglik(model)= -284.5   Loglik(intercept only)= -501.5
    Chisq= 433.89 on 2 degrees of freedom, p= 6.1e-95
Number of Newton-Raphson Iterations: 5
n= 1000
```

```r
# recover qis: these are medians!
predictions(fit,
            variables = list(x1 = "minmax"),
            newdata   = datagrid(x2 = "Label 2"))
```

```
      x2 Estimate Std. Error    z Pr(>|z|)     S 2.5 % 97.5 %
 Label 2    0.306    0.00899 34.0   <0.001 841.0 0.288  0.324
 Label 2    1.654    0.07073 23.4   <0.001 399.1 1.515  1.792

Type: response
```

```r
comparisons(fit,
            variables = list(x1 = "minmax"),
            newdata   = datagrid(x2 = "Label 2"))
```

```
      x2 Estimate Std. Error     z Pr(>|z|)     S 2.5 % 97.5 %
 Label 2     1.35      0.072 18.7   <0.001 257.0  1.21   1.49

Term: x1
Type: response
Comparison: Max - Min
```

```r
# predict is returning median(Y) not E(Y) here
grid <- tibble(x1 = c(min(x1), max(x1)), x2 = "Label 2") |>
  glimpse()
```

```
Rows: 2
Columns: 2
$ x1 <dbl> -1.404887, 1.620520
$ x2 <chr> "Label 2", "Label 2"
```

```r
predict(fit, type = "response", newdata = grid)
```

```
       1        2
0.305887 1.653676
```

```r
median_hi; median_lo
```

```
[1] 1.77236
```

```
[1] 0.2885369
```

```r
mean_hi; mean_lo
```

```
[1] 1.919975
```

```
[1] 0.3125683
```

```r
# try with flexsurv
fit_fs <- flexsurvreg(Surv(time, status) ~ x1 + x2, data = data, dist = "lnorm")
predict(fit_fs, type = "response", newdata = grid)
```

```
# A tibble: 2 x 1
  .pred_time
       <dbl>
1      0.330
2      1.79
```

Exercise 2 Claims and CIs

Read Rainey (2014) and McCaskey and Rainey (2015). These papers describe in detail my advice to "only make a claim if the claim holds for the *entire* confidence interval."

Give two examples (see below) of how current practice deviates from this advice. Explain a how my advice applies to these two situations. Explain why these deviations from my advice matter (or not).

Examples:

1. McCaskey and Rainey (2015): Claiming "substantively significant" if the point estimate is meaningful (and statistically significant).
2. Rainey (2014): Claiming "no effect" when an estimate is not statistically significant.

Exercise 3 Power

Background

Suppose you are conducting a simple, balanced treatment-control experiment. You know that the SD of the outcome will be about $\widetilde{SD}$. Rainey (2025) shows that the SE of the estimated effect will be about $\frac{2 \cdot \widetilde{SD}}{\sqrt{2 \cdot n}}$, where n is the number of participant *per condition*. Further, the minimum detectable effect with 95% power is $3.3 \cdot SE$.

Details of the Experiment

- Your outcome is affective polarization, measured as a 101-point feeling thermometer directed at the out-party ("Republicans" or "Democrats"). This outcome typically has an SD of about 25 in surveys like the ANES or those conducted on crowd-sourced platforms like MTurk.
- Your treatment a two-minute video showing Republicans and Democrats enjoying a beer together.

If you have a simple, balanced treatment-control experiment that you'd rather write about, feel free to replace the details above with the analogous information.

Questions

To answer these question, you can refer to Rainey (2025) if helpful–there are lots of useful rules there. But the *Background* above is sufficient!

1. For the treatment and outcome above, what is the smallest treatment effect that is substantively meaningful. Alternatively, what is the largest treatment effect that is substantively trivial. Come up with a precise number on the 101-point feeling thermometer scale.[1]
2. For samples of size 100, 200, 500, and 2,000 *per condition*, what is the minimum detectable effect with 95% power?
3. Suppose you are applying for grant funding for this experiment. Respondents cost $2.00 each. Write a paragraph justifying a specific number of respondents. Describe the treatment effects you are targeting (i.e., the smallest substantively meaningful effect), the power target (e.g., 80%, 90%, or 95%), and the sample size to obtain that power.

Exercise 4 Literatures

Coming soon.

References

McCaskey, Kelly, and Carlisle Rainey. 2015. "Substantive Importance and the Veil of Statistical Significance." *Statistics, Politics and Policy* 6 (1-2). https://doi.org/10.1515/spp-2015-0001.

Rainey, Carlisle. 2014. "Arguing for a Negligible Effect." *American Journal of Political Science* 58 (4): 1083–91. https://doi.org/10.1111/ajps.12102.

———. 2025. "Power Rules: Practical Advice for Computing Power (and Automating with Pilot Data)." http://dx.doi.org/10.31219/osf.io/5am9q_v2.

[1]For example, a 5-point treatment effect might mean that the average in the treatment group is 55 and the average in the control group is 50.